

Estimation of conduct and demand: evidence from a Norwegian cement cartel

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1 Introduction

Classic industrial organization theory reasons the fall of cartels by an incentive to cheat. A seminal work by (Green and Porter 1984) shows that a non-monitored cartel generates endogenously cycles of price wars from punishment of overproduction. However, even if this pattern has been replicated with constrained-quantity competition markets, showing it empirically is a complex endeavor due to data limitations. This paper exploits a Norwegian cement cartel case in order to estimate demand and conduct parameters. Given that this cartel went through all three cases (i.e., public cartel, public monopoly, competition), it provides meaningful insights into the formation and end of a cartel in a homogeneous-good constrained market. We revisit both the theoretical and empirical strategies of the original paper (Röller and Steen 2006), then update some results. Section 2 presents the cartel background and a simple theoretical modelization. Section 3 estimates inverse demand and conduct. Section 4 concludes.

2 Background

2.1 Norwegian Cement Cartel

At the end of World War I, the capacity of the three major cement manufacturers was double the demand. This created a price war, and in 1923, the common sales office was established to centralize production. This cartel relied on a sharing rule that allocated profit proportionally to the capacity. This rule was inherently inefficient, as it incentivized capacity increases. In 1968, the cartel collapsed, and all firms merged to form a monopoly thereafter.

This case manifests three interesting results that are of interest: i) the sharing rule created an incentive to overproduce and export; ii) the expected merger welfare outpassed the cartel welfare, which indicates inefficiency; iii) competition would have generated higher surpluses than cartel or merger.

2.2 Model

Let N domestic cement firms (e.g, $N = 3$ in this case) with a linear inverse demand function $P_t = a_t - bQ_t^D$. As in the original paper, we assume that the export price R_t is taken as given, due to competitiveness of the world cement industry. Each firm has a given capacity K_{it} and can sell domestically or abroad without further costs (we assume no outward trade policies) such as $q_{it} = q_{it}^D + q_{it}^X \leq K_{it}$. The individual firm profit function writes as:

$$\pi_{it} = P_t q_{it}^D + R_t q_{it}^X - c q_{it}$$

with c marginal cost. We assume no import.¹ Furthermore, we adopt Roller and Steen's findings that $R_t < c$

2.2.1 Case 1: Cartel with sharing rule

Norway Cement Cartel had a sharing rule based on capacity. The share of profit of a firm s_{it} was decided by a following rule:

$$s_{it} = \frac{K_{it}}{\sum_j K_{jt}}$$

with K_{it} capacity of firm i . Excess domestic stock is exported at price R_t , that is $q_{it}^X = K_{it} - q_{it}^D$. Profit function of firm i writes as:

$$\pi_{it} = P_t s_{it} Q_t^D + R_t (K_{it} - q_{it}^D) - c K_{it}$$

As in the original paper, we proceed first by determining optimal aggregate quantity Q_t^D then idiosyncratic quantity q_{it} . It results that:

$$\max_{Q_t^D} \Pi_t = P_t(Q_t^D) Q_t^D + R_t \left(\sum_i K_{it} - Q_t^D \right) - c \sum_i K_{it}$$

$$P_t'(Q_t^D) Q_t^D + P_t(Q_t^D) - R_t = 0$$

¹Following the assumption of Roller and Steen, based on the fact that there were very little import of cement.

2.2.2 Case 2: Monopoly

Under monopoly after merging, the firm chooses the quantity maximizing its profit. Since we assume no import and export is always a loss for a firm, we have a typical firm's problem:

$$\max_{Q_t^M} \Pi_t = P_t(Q_t^M)Q_t^M - cQ_t^M$$

The FOC of the firm is

$$P_t'(Q_t)Q_t + P(Q_t) = c$$

2.2.3 Case 3: Cournot competition benchmark

Roller and Steen also present the hypothetical case of Cournot competition as a benchmark. Assuming symmetric structure across the N firms, the maximization problem is:

$$\max_{Q_t^C} \Pi_t = \frac{1}{N} P_t(Q_t^C)Q_t^C - cQ_t^C$$

Thus FOC in this case is

$$\frac{1}{N} P_t'(Q_t)Q_t + P(Q_t) = c$$

3 Estimation strategy

Further summary statistics are given in [Table 1](#), [Table 2](#), and [Table 3](#). We observe that: i) Total production increases strongly post-merger; ii) the merger coincides with higher prices and export reliance; iii) input prices are stable across the period.

3.1 Estimation of Demand

Given the homogeneity of cement, product differentiation is unlikely to play an important role in demand. The authors therefore estimate a relatively parsimonious inverse demand function. To address the endogeneity of quantity, they employ cost shifters (e.g, labor, material, and energy prices) as instruments. One important detail is that the the authors additionally includes

the export price as an instrument, although only three cost shifters are explicitly discussed in the paper. We follow the empirical implementation of the original study.

3.1.1 Specification

To account for sluggish adjustment in the construction industry, we estimate an Autoregressive Distributed Lag (ADL) specification. The baseline inverse demand equation is:

$$p_{n,t} = \beta_0 + \beta_1 q_{n,t} + \beta_2 q_{n,t-1} + \beta_3 p_{n,t-1} + \gamma_1 \text{byan}_t + \gamma_2 \text{byan}_{t-1} + \varepsilon_t$$

where **byan** denotes the Norwegian building activity index.

Another concern is that the lag structure is not explicitly justified in the original paper. We may indeed expect that: i) cement is bought on the spot, which makes previous values not relevant; ii) in contrary, cement is a persistent good that shows stability (i.e, due to norms and other fixed costs that makes prices stable over time). We consider then a model that address both issues.

$$p_{n,t} = \beta_0 + \beta_1 q_{n,t} + \gamma_1 \text{byan}_t + \varepsilon_t$$

$$p_{n,t} = \beta_0 + \beta_1 q_{n,t} + \beta_2 q_{n,t-1} + \beta_3 q_{n,t-2}$$

$$+ \alpha_1 p_{n,t-1} + \alpha_2 p_{n,t-2} + \gamma_1 \text{byan}_t + \gamma_2 \text{byan}_{t-1} + \gamma_3 \text{byan}_{t-2} + \varepsilon_t$$

Finally, the original paper pools cartel and monopoly periods when estimating demand, implicitly assuming that the underlying demand relationship remains stable over time. However, the 1968 merger may have altered market dynamics. Under monopoly, incentives to overproduce disappear, potentially affecting the relationship between domestic quantity and price.

$$p_{n,t} = \beta_0 + \beta_1 q_{n,t} + \beta_2 \text{post1968}_t + \beta_3 (q_{n,t} \times \text{post1968}_t) + \Gamma X_t + \varepsilon_t$$

where X_t includes lagged prices, lagged quantities, and the building activity index.

3.1.2 Results

All regression results are displayed in [Table 7](#). The IV regression with the ADL(1,1) specification (Panel 3) successfully replicates the coefficient magnitude reported in ([Röller and Steen 2006](#)). Consistent with economic theory, domestic demand is significantly downward sloping,

and the estimated quantity coefficient is remarkably close to the original paper. Domestic prices also exhibit substantial persistence, as indicated by the significance of the lagged price term.

However, first-stage diagnostics indicate a weak-instrument concern. The excluded instrument test yields a relatively low first-stage F-statistic, suggesting limited identification strength. This does not invalidate the estimates, but implies that inference should be interpreted with caution.

To assess whether the baseline ADL(1,1) specification is sufficient, we conduct lag-order diagnostics using the Breusch–Godfrey (BG) test together with information criteria (AIC/BIC). Results in [Table 4](#) and [Table 5](#) suggest that the baseline specification may be restrictive, particularly for domestic consumption. While domestic prices and the building activity index do not exhibit substantial residual autocorrelation at lag one, the BG test strongly rejects the null of no serial correlation for consumption at lag one ($p = 0.01$). This issue disappears from lag two onward.

Moreover, both AIC and BIC decline monotonically as lag length increases, suggesting that richer dynamic structures may improve fit. This raises the possibility that weak quantity effects in the baseline model partly reflect dynamic misspecification rather than solely weak identification.

We therefore estimate richer dynamic specifications, including ADL(2,2) and partial-lag alternatives. However, these specifications substantially weaken coefficient stability (Panel 4) fail overidentification diagnostics and overall significance. Given the small sample size and strong serial dependence among lagged regressors, richer specifications likely suffer from overfitting and multi colinearity. We therefore retain the parsimonious ADL(1,1) specification as the preferred baseline.

Separated model is presented in Panel 5. The interaction term between quantity and the post-1968 period is statistically significant, suggesting a change in the relationship between domestic quantity and prices after the merger.

However, the implied monopoly-period demand slope becomes positive, which is economically implausible for a homogeneous good market. This suggests that the estimated structural

break should be interpreted cautiously. Given the limited number of post-merger observations and the expanded instrument set required for the interaction specification, the result likely reflects finite-sample instability and over parameterization rather than a genuine reversal of demand behavior.

Accordingly, while the structural-break specification suggests heterogeneity across periods, the baseline pooled specification remains economically more plausible.

3.2 Evaluation of conduct

3.2.1 Specification

From the model presentation in section 2, we can observe that the three models yield first-order conditions of similar form. Specifically, the FOC of our model with a common sales office and export yields has world price R playing the role of marginal cost. Conversely, both FOCs of Cournot model and monopoly case have marginal cost on the right-hand side. However, the Cournot model has $1/3$ multiplied by the markup term, as we have three firms in the Norwegian cement industry case. This suggest that we need to compare between three specifications.

$$P'_t(Q_t^D)Q_t^D + P(Q_t^D) = R_t$$

$$\frac{1}{3}P'_t(Q_t)Q_t + P(Q_t) = c$$

$$P'_t(Q_t)Q_t + P(Q_t) = c$$

Let a AR(1) demand model defined as

$$p_t = \rho p_{t-1} + \beta_{sr} q_t + \beta_{lr} q_{t-1} + \varepsilon_t$$

such that, if we consider a permanent increase in quantity as $\Delta q_t = \Delta q_{t+1} = 1 \ \forall t$,

$$\Delta p_T = (\beta_{sr} + \beta_{lr}) + \rho(\beta_{sr} + \beta_{lr}) + \rho^2(\beta_{sr} + \beta_{lr}) + \dots = \frac{1}{1 - \rho}(\beta_{sr} + \beta_{lr})$$

assuming that $|\rho| < 1$. We have the marginal revenue of a linear inverse demand function defined as $c = P - \beta Q$, for $\beta = \frac{\partial P}{\partial Q}$. We then estimate a markup value as:

$$\mu_{sr} = -q \times \beta_{sr}$$

and

$$\mu_{lr} = -q \times \beta_{lr}$$

We may have multiple proxies for marginal cost: i) export price of Norwegian cement; ii) short-run marginal cost derived from cost of labor and production; iii) estimates of marginal cost from the theoretical model presented in section 2. Considering the lack of observations, we first rely on the pooled 2SLS for this estimation. Estimates with the separated model are also computed for sake of comparison.

Let a simple pricing model as: $p_t = c + \mu q_t + \varepsilon_t$ that can be estimated through OLS given that we have strong proxies of marginal cost. A priori, we should expect a conduct parameter ranging between 1/3 and 1 for the cartel period (considering the cartel was less strong in markup than the monopoly) and equal to 1 for the monopoly period.

3.2.2 Results

For the cartel era (e.g, before 1968), estimates are computed in [Table 8](#), [Table 9](#), and [Table 10](#). For the monopoly era, as we do have only two proxies remaining, estimates are in [Table 11](#) and [Table 12](#).

The conduct estimates provide evidence of substantial market power during the cartel period, although results remain sensitive to the choice of marginal-cost proxy. Specifications using export prices and predicted marginal costs generally yield positive and statistically significant markup coefficients, consistent with coordinated pricing behavior and rejection of competitive conduct. By contrast, estimates based on short-run average cost display substantial instability and occasional sign reversals, suggesting that this proxy poorly captures marginal cost. Overall, long-run markup coefficients in the non-constant OLS yield the most convincing results.

Monopoly-period estimates are weaker and less stable overall, likely reflecting both the limited post-merger sample size and reduced variation in conduct after the 1968 merger. Taken together, the results support the existence of significant market power under the Norwegian

cement cartel, while also highlighting the difficulty of precisely identifying conduct in a small-sample setting with unobserved marginal costs.

Nonetheless, we should expect that export prices and short-run average costs are lower than true marginal cost, given the overproduction of cartel firms for the former and the obvious lack of source of costs for the latter, thus understating the true conduct parameter value.

4 Conclusion

We complete the paper of ([Röller and Steen 2006](#)) by allowing for more specifications and completing concerns that were unanswered. We identify multiple issues that lowers the scope of interpretations of estimates, the first one being the lack of observations. Results are relatively consistent across models for the estimation of inverse demand. In the other hand, even though estimates for conduct are not implausible, we may still fear a clear identification issue because of approximate proxies for marginal cost.

References

- Green, Edward J., and Robert H. Porter. 1984. "Noncooperative Collusion under Imperfect Price Information." *Econometrica* 52 (1): 87. <https://doi.org/10.2307/1911462>.
- Röller, Lars-Hendrik, and Frode Steen. 2006. "On the Workings of a Cartel: Evidence from the Norwegian Cement Industry." *The American Economic Review* 96 (1): 321–38. <http://www.jstor.org/stable/30034368>.

Appendix

Table 1. Summary statistics for the pre-merger period.

Variable	Mean	SD	Min	Max
Total production	1397665	452575.31	799878	2297088
Domestic price	459.49	72.18	359.08	594.76
Total exports	227745.33	247054.1	800	840596
Export price	393.92	107.1	235.07	641.16
Price of materials	99.23	19.31	63.37	121.73
Price of electric- ity	100.66	49.06	50.06	185.82
Real wage	96337.91	15128.46	69173	116776.6
Domestic con- sumption	1128108.58	232149.24	740249	1539344

Table 2. Summary statistics for the post-merger period.

Variable	Mean	SD	Min	Max
Total production	2425886.86	328250.14	1786000	2740169
Domestic price	593.37	90.36	409.72	706.29
Total exports	538526.33	456582.76	0	1217277
Export price	255.99	38.99	227.54	373.19
Price of materials	153.24	42.65	71.96	208.38
Price of electric- ity	99.88	35.84	50.82	150.23
Real wage	167627.40	39990.56	115812.92	211595.31
Domestic con- sumption	1631846.86	110257.77	1403078	1795089

Table 3. Summary statistics for the full sample.

Variable	Mean	SD	Min	Max
Total production	1911775.93	651611.24	799878	2740169
Domestic price	524.13	105.03	359.08	706.29
Total exports	409034.25	409493.55	0	1217277
Export price	327.33	106.59	227.54	641.16
Price of materials	125.30	42.25	63.37	208.38
Price of electricity	100.28	42.43	50.06	185.82
Real wage	130753.52	46597.23	69173	211595.31
Domestic consumption	1371292.57	313395.02	740249	1795089

Table 4. Table of lag order autocorrelation for price.

Lag order	AIC	BIC	BG p-value	adj. R^2
1	311.14	315.14	0.14	0.7
2	300.91	306.09	0.47	0.71
3	291.8	298.09	0.24	0.71
4	282.25	289.56	0.67	0.71
5	272.12	280.37	0.15	0.72
6	260.53	269.62	0.3	0.75

Table 5. Table of lag order autocorrelation for demand index.

Lag order	AIC	BIC	BG p-value	adj. R^2
1	317.22	321.22	0.95	0.98
2	308.73	313.92	0.47	0.97
3	298.56	304.85	0.42	0.97
4	286.76	294.07	0.66	0.97
5	278.49	286.73	0.3	0.97
6	267.42	276.51	0.24	0.97

Table 6. Table of lag order autocorrelation for consumption.

Lag order	AIC	BIC	BG p-value	adj. R^2
1	725.09	729.09	0.01	0.9
2	693.53	698.72	0.42	0.91
3	670.65	676.94	0.58	0.89
4	646.64	653.96	0.26	0.87
5	623.17	631.42	0.58	0.84
6	598.53	607.61	0.42	0.82

Table 7. Two-Stage Least Squares regression table results of inverse demand

	<i>Dependent variable:</i>				
	pnr				
	(1)	(2)	(3)	(4)	(5)
qn	−0.0003*** (0.0001)	−0.001** (0.0004)	−0.001** (0.0004)	0.0001 (0.0003)	−0.001*** (0.0002)
post1968					−934.966** (430.492)
byan	0.373*** (0.065)	−0.148 (0.369)	−0.190 (0.297)	0.060 (0.236)	0.098 (0.198)
pnr_lag			0.532* (0.259)	0.305 (0.292)	
pnr_lag2				0.480 (0.376)	
qn_lag		0.001* (0.0004)	0.001** (0.0003)	0.0001 (0.0002)	0.0004** (0.0002)
byan_lag		0.482 (0.363)	0.334 (0.300)		0.076 (0.205)
qn_lag2				0.00003 (0.0002)	
byan_lag1				−0.301 (0.334)	
byan_lag2				0.203 (0.260)	
qn:post1968					0.001** (0.0003)
Constant	388.009*** (64.617)	451.367*** (109.109)	239.361* (135.189)	−180.004 (182.022)	741.716*** (101.969)
Observations	29	28	28	23	28
R^2	0.570	0.232	0.527	0.843	0.800
Adjusted R^2	0.537	0.098	0.420	0.753	0.743
Residual Std. Error	71.441 (df = 26)	101.552 (df = 23)	81.450 (df = 22)	58.303 (df = 14)	54.212 (df = 21)

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 8. Estimation of conduct parameter in cartel era, using export price proxy

	<i>Dependent variable:</i>					
	pnr		gap_pexpr		pnr (no const.)	
	(1)	(2)	(3)	(4)	(5)	(6)
pexpr	0.339 (0.270)	0.339 (0.270)			0.983 *** (0.067)	0.983 *** (0.067)
markup_sr	−0.218 (0.127)		0.080 *** (0.009)		0.086 *** (0.026)	
markup_lr		−0.640 (0.371)		0.235 *** (0.027)		0.252 *** (0.075)
Constant	543.517 ** (222.492)	543.517 ** (222.492)				
Observations	14	14	14	14	14	14
R^2	0.854	0.854	0.851	0.851	0.995	0.995
Adjusted R^2	0.827	0.827	0.839	0.839	0.994	0.994
Residual Std. Error	30.436 (df = 11)	30.430 (df = 11)	34.867 (df = 13)	34.867 (df = 13)	36.192 (df = 12)	36.192 (df = 12)
F Statistic	32.063 ***	32.063 ***	74.033 ***	74.033 ***	1,129.367 ***	1,129.367 ***
<i>Note:</i>				* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$		

Table 9. Estimation of conduct parameter in cartel era, using authors predictions as proxy

	<i>Dependent variable:</i>					
	pnr		gap_pmc		pnr (no const.)	
	(1)	(2)	(3)	(4)	(5)	(6)
predict- edmc	0.941 ***	0.941 ***			0.968 ***	0.968 ***
	(0.172)	(0.172)			(0.029)	(0.029)
markup_sr	0.037		0.036 ***		0.049 ***	
	(0.078)		(0.004)		(0.012)	
markup_lr		0.107		0.106 ***		0.143 ***
		(0.229)		(0.013)		(0.036)
Constant	23.580	23.580				
	(147.228)	(147.228)				
Observa- tions	14	14	14	14	14	14
R^2	0.955	0.955	0.842	0.842	0.999	0.999
Adjusted R^2	0.947	0.947	0.829	0.829	0.999	0.999
Residual	16.871	16.871	16.315	16.315	16.171	16.171
Std. Error	(df = 11)	(df = 11)	(df = 13)	(df = 13)	(df = 12)	(df = 12)
F Statistic	116.760 ***	116.760 ***	69.085 ***	69.085 ***	5,680.915 ***	5,680.915 ***
<i>Note:</i>				* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$		

Table 10. Estimation of conduct parameter in cartel era, using short-run average cost as proxy

	<i>Dependent variable:</i>					
	pnr		gap_sac		pnr (no const.)	
	(1)	(2)	(3)	(4)	(5)	(6)
shortAC	0.103 *** (0.032)	0.103 *** (0.032)			0.175 *** (0.009)	0.175 *** (0.009)
markup_sr	−0.045 (0.105)		−0.920 *** (0.188)		0.199 *** (0.014)	
markup_lr		−0.132 (0.309)		−2.697 *** (0.551)		0.583 *** (0.041)
Constant	347.685 ** (149.077)	347.685 ** (149.077)				
Observations	14	14	14	14	14	14
R^2	0.914	0.914	0.648	0.648	0.997	0.997
Adjusted R^2	0.899	0.899	0.621	0.621	0.997	0.997
Residual Std. Error	23.282 (df = 11)	23.282 (df = 11)	704.455 (df = 13)	704.455 (df = 13)	27.250 (df = 12)	27.250 (df = 12)
F Statistic	58.695 ***	58.695 ***	23.982 ***	23.982 ***	1,996.684 ***	1,996.684 ***
<i>Note:</i>				* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$		

Table 11. Estimation of conduct parameter in monopoly era, using export price as proxy

	<i>Dependent variable:</i>					
	pnr	gap_pexpr	gap_pexpr	pnr		
	(1)	(2)	(3)	(4)	(5)	(6)
pexpr	0.148 (0.756)	0.148 (0.756)			0.134 (0.732)	0.134 (0.732)
markup_sr	0.256 (0.313)		0.242 *** (0.018)		0.401 ** (0.136)	
markup_lr		0.750 (0.918)		0.708 *** (0.052)		1.175 ** (0.398)
Constant	199.208 (385.442)	199.208 (385.442)				
Observations	14	14	14	14	14	14
R^2	0.091	0.091	0.935	0.935	0.980	0.980
Adjusted R^2	−0.074	−0.074	0.930	0.930	0.977	0.977
Residual	93.647	93.647	92.118	92.118	90.742	90.742
Std. Error	(df = 11)	(df = 11)	(df = 13)	(df = 13)	(df = 12)	(df = 12)
F Statistic	0.552 (df = 2; 11)	0.552 (df = 2; 11)	187.587 *** (df = 1; 13)	187.587 *** (df = 1; 13)	299.766 *** (df = 2; 12)	299.766 *** (df = 2; 12)
<i>Note:</i>				* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$		

Table 12. Estimation of conduct parameter in monopoly era, using short-run average cost as proxy

	<i>Dependent variable:</i>					
	pnr		gap_sac	gap_sac	pnr	
	(1)	(2)	(3)	(4)	(5)	(6)
shortAC	−0.014 (0.228)	−0.014 (0.228)			0.015 (0.214)	0.015 (0.214)
markup_sr	0.288 (0.281)		−0.099 *** (0.028)		0.418 *** (0.114)	
markup_lr		0.844 (0.823)		−0.289 *** (0.081)		1.224 *** (0.333)
Constant	202.494 (398.043)	202.494 (398.043)				
Observations	14	14	14	14	14	14
R^2	0.088	0.088	0.493	0.493	0.980	0.980
Adjusted R^2	−0.077	−0.077	0.454	0.454	0.977	0.977
Residual Std. Error	93.793 (df = 11)	93.793 (df = 11)	145.127 (df = 13)	145.127 (df = 13)	90.851 (df = 12)	90.851 (df = 12)
F Statistic	0.533 (df = 2; 11)	0.533 (df = 2; 11)	12.622 *** (df = 1; 13)	12.622 *** (df = 1; 13)	299.036 *** (df = 2; 12)	299.036 *** (df = 2; 12)
<i>Note:</i>				* $p < 0.1$; * * $p < 0.05$; * * * $p < 0.01$		